



Designing and Building AI Products and Services

Module 1 Office Hours
Bruce Lam



Agenda

01

4 Stages & 8 choices

Unpack & clarify select stages

02

Case in Practice

03

LLM Limitations

04

EMcode/notebooks

Notebook Walkthrough

ML vs. AI

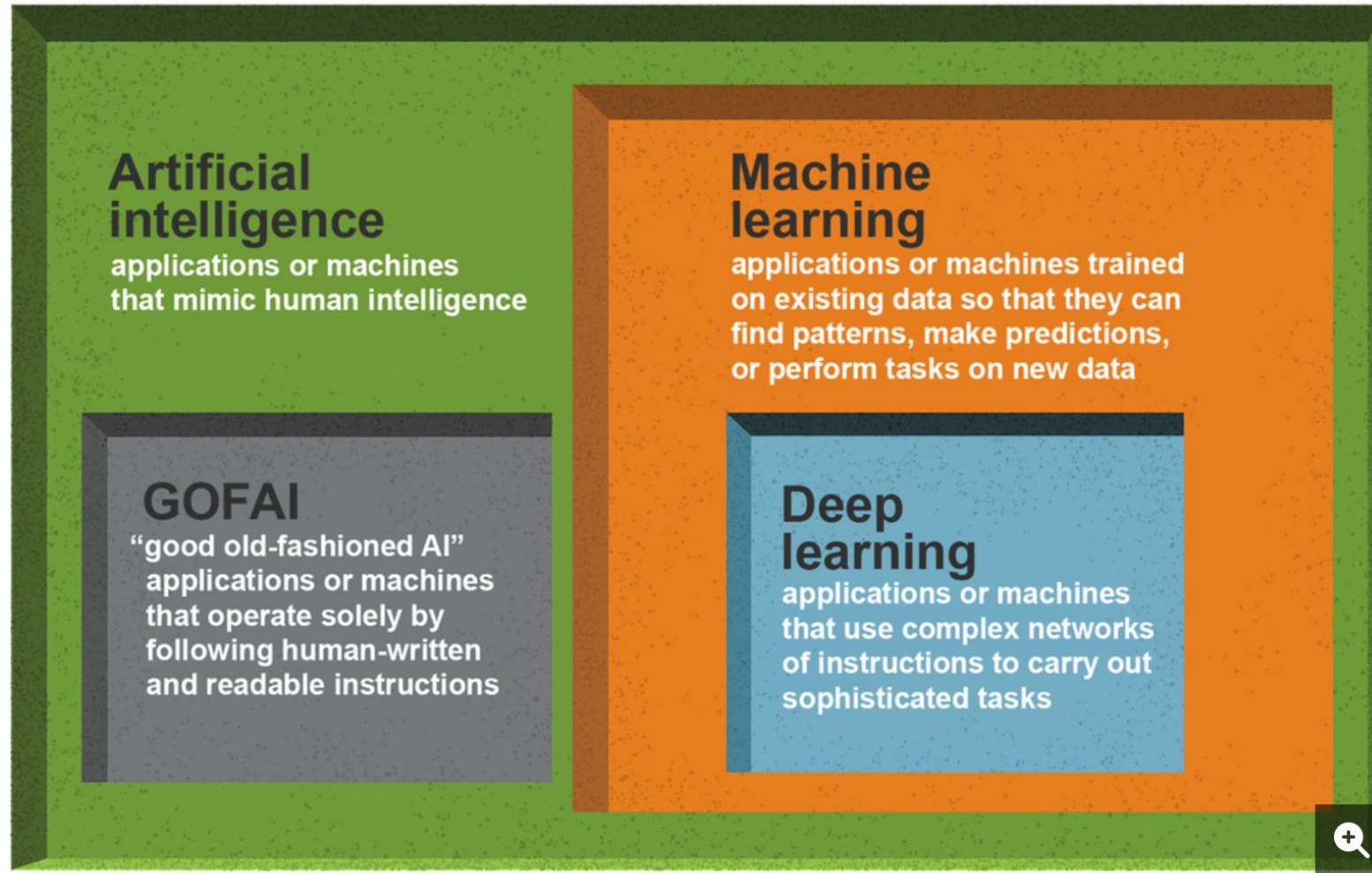
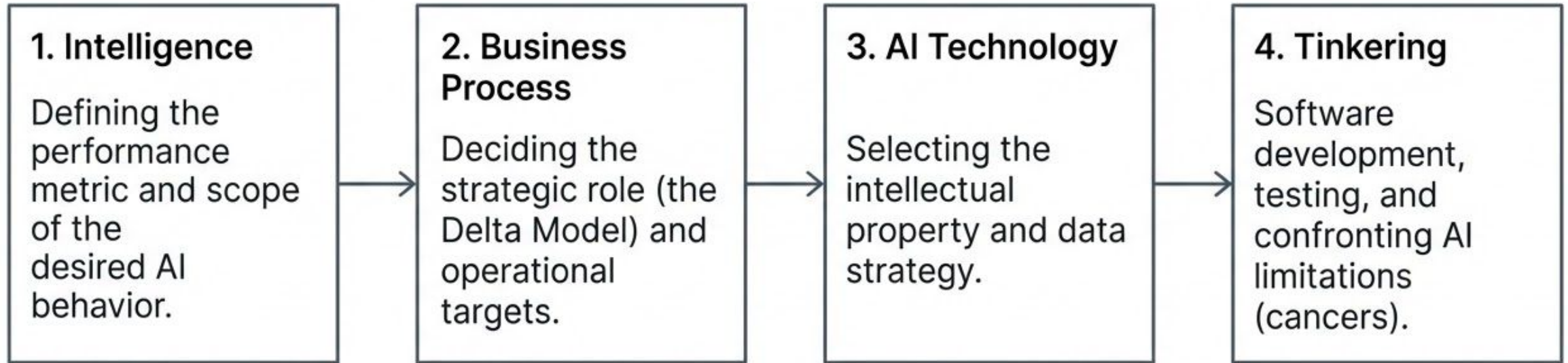


Diagram explaining the difference between categories of artificial intelligence

Credit: Lance Hayashida for Caltech Science Exchange

<https://scienceexchange.caltech.edu/topics/artificial-intelligence-research/artificial-intelligence-vs-machine-learning>

The AI Design Process Blueprint



Key Takeaway: The process is rarely a straight line. **Hitting a limitation** in **Stage 4** requires **pivoting back** to **Stages 1 and 2**.

The Master Architecture: 4 Stages, 8 Choices

A framework for designing AI products, navigating four sequential stages, each demanding two critical strategic choices.

Prerequisites: A clearly defined and understood problem, and executive sponsorship (for corporate situations) are essential before applying this framework.

Stage 1: Intelligence

Define the required intelligence and behavior before introducing AI. Focus on the problem, not the technology.

Node 1: Performance Metric



Define a quantifiable goal and establish a baseline for measuring success. What specific metric will demonstrate improvement?

Node 2: Scope

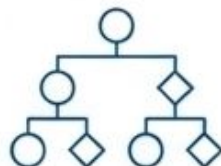


Clearly define the boundaries and context of the AI's operation. What is included in the system's purview, and what is excluded?

Stage 2: Business Process

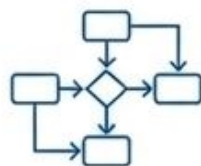
Integrate the AI system into the existing or new business workflow. Align AI capabilities with operational needs.

Node 3: Strategic Choice



Decide on the strategic impact. Will the AI automate existing tasks, augment human capabilities, or create entirely new processes?

Node 4: Operational Target



Identify the specific process or workflow that the AI will improve or transform. Where will the AI be deployed?

Stage 3: AI Technology

Select the appropriate AI tools, models, and data strategies. The technology must serve the intelligence goal.

Node 5: IP Approach



Determine the strategy for intellectual property. Will you build proprietary models, buy commercial solutions, or leverage open-source?

Node 6: Data Strategy



Plan for data acquisition, cleaning, labeling, and governance. How will you ensure a continuous supply of high-quality data?

Stage 4: Tinkering

Iterate, test, and refine the AI system for real-world deployment. Continuous improvement is key.

Node 7: Software Development & MLOps



Develop the necessary software infrastructure, APIs, and user interfaces to support the AI model in production, including robust MLOps for continuous experimentation and model lifecycle management.

Node 8: Mitigating Cancers



Proactively identify and address potential risks like bias, model drift, security vulnerabilities, and ethical concerns.

Choice 1: The Performance Metric

Identify Baseline

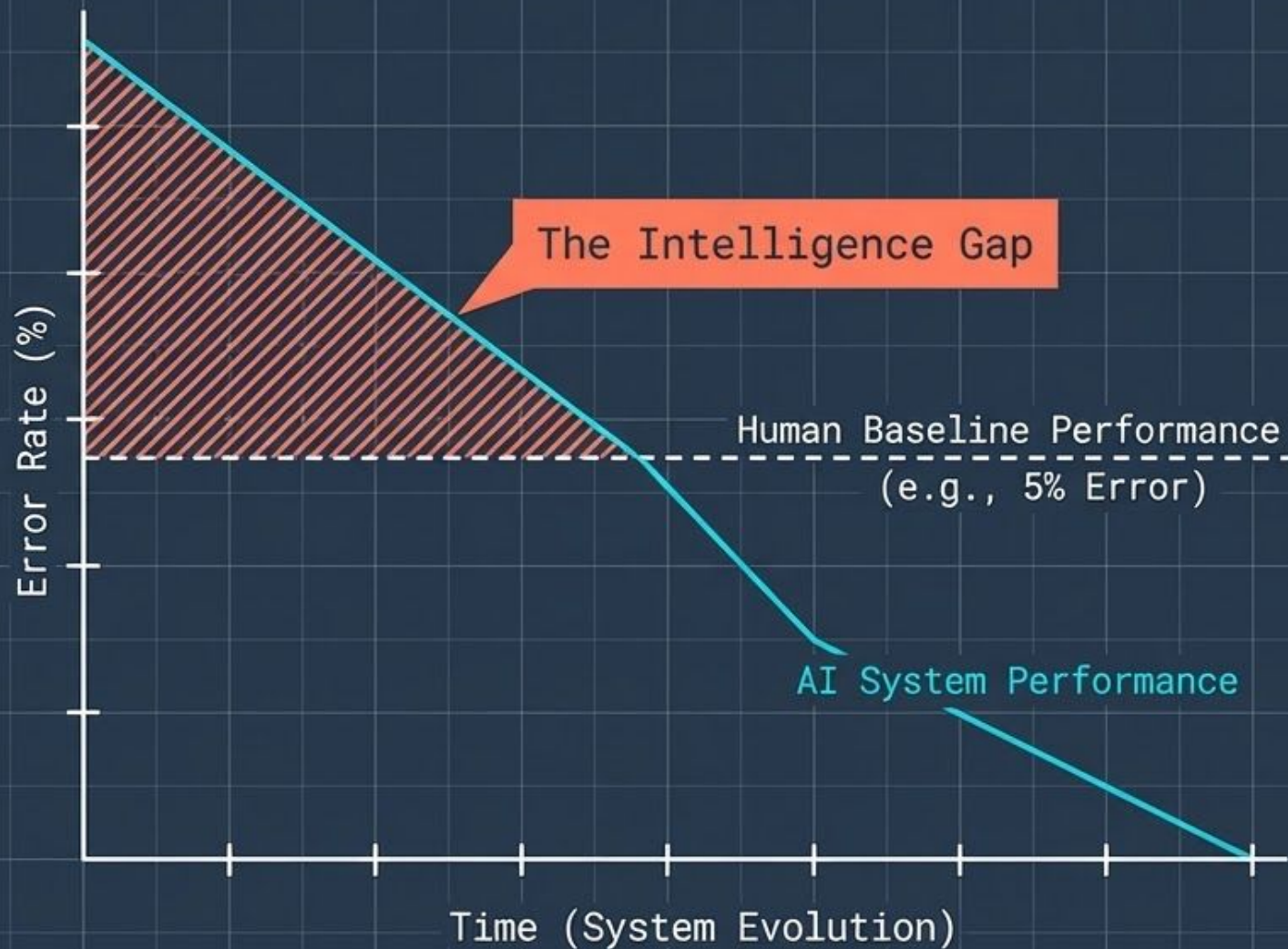
Establish the current human baseline error rate for the specific task.

Set Quantifiable Targets

Define the exact mathematical threshold required to close the intelligence gap.

Continuous Calibration

Revisit targets dynamically as the technological frontier shifts.

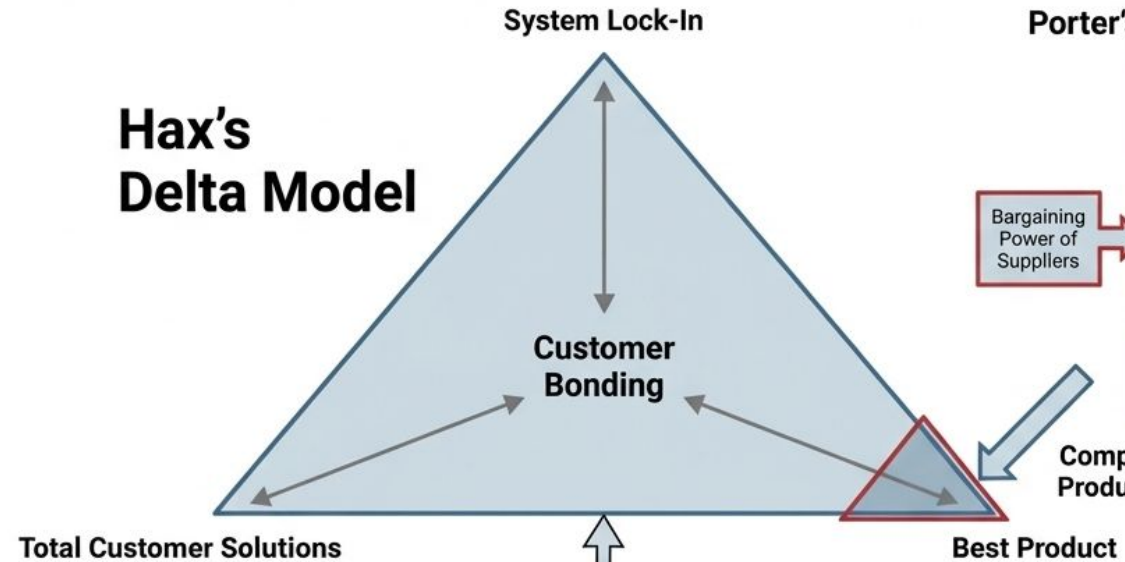


Stage 2: Operational Targets

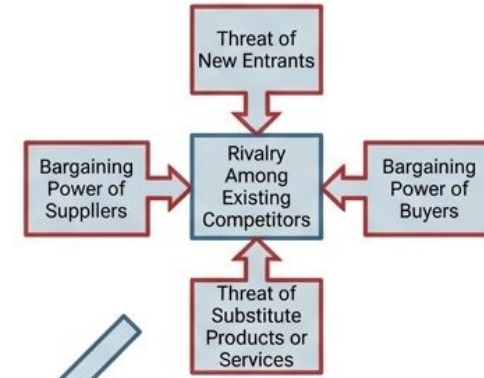
Focus Area	Stage 1: Intelligence Performance Metrics	Stage 2: The Business Target
Customer Support	Metric: 95% NLP accuracy for answering FAQs.	Target: Lower uncompleted calls by 80% while keeping costs flat.
OpEx / Efficiency	Metric: Model processes 1,000 tickets per minute.	Target: Slash total call center costs by 70% at current service levels.
Legal / Compliance	Metric: High-fidelity translation of technical terms.	Target: Cut document translation costs by 80% via "Review-Only" workflows.
Product UX	Metric: Voice-to-text error rate under 1%.	Target: 100% keyboard-free flight booking via automated voice workflows.

Stage 2: Strategic Choice

Hax's Delta Model



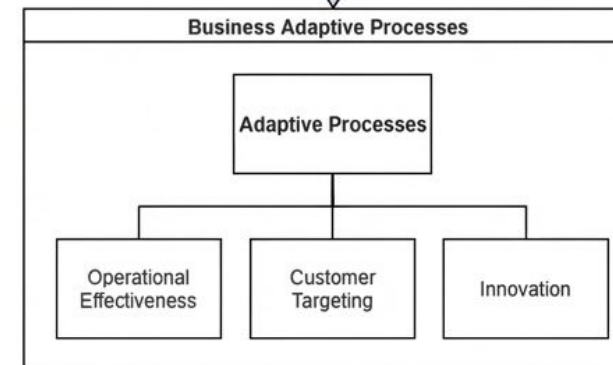
Porter's 5 Forces Model



Competitive Analysis for Product-Centric Strategy

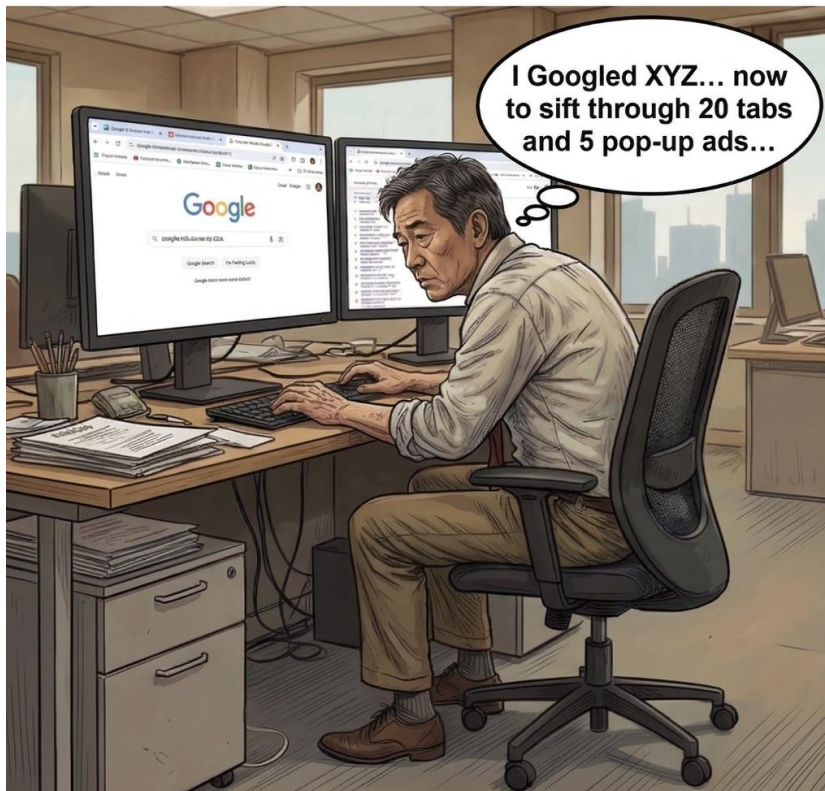
The 10 'Haxioms' of Strategy in the AI Era

1. The center of strategy is the customer.
2. You don't win by beating the competition; you win by achieving customer bonding.
3. Strategy is not war; it is love.
4. Include customers, suppliers, and complementors as key constituencies.
5. Understand your customer deeply. Strategy is done one customer at a time.
6. Commodities only exist in the minds of the inept.
7. Strategy relies on customer segmentation and bundling unique competencies.
8. Reject truisms like "the customer is always right".
9. Strategic planning is a dialogue seeking consensus on organizational direction.
10. Metrics are essential; experimentation is crucial.



Stage 3: Technology

Me, an Elder Millennial/Gen X, preparing for a 30-minute research deep dive...



My Gen Z intern, 8 seconds later:



"Oh, I already asked Chat. It made a bulleted list. I just Slacked it to you."

Stage 3: AI Technology



Berkeley Function-Calling Leaderboard

BFCL: From Tool Use to Agentic Evaluation of Large Language Models

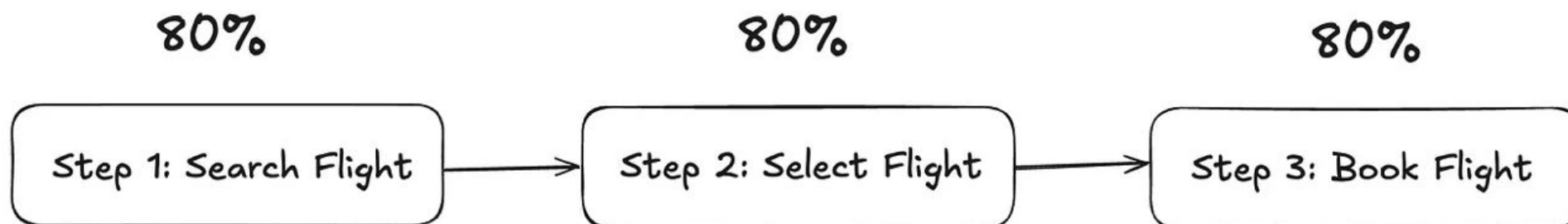
The Berkeley Function Calling Leaderboard (BFCL) V4 evaluates the LLM's ability to call functions (aka tools) accurately. This leaderboard consists of real-world data and will be updated periodically. For more information on the evaluation dataset and methodology, please refer to our blogs: [BFCL-v1](#) introducing AST as an evaluation metric, [BFCL-v2](#) introducing enterprise and OSS-contributed functions, [BFCL-v3](#) introducing multi-turn interactions, and [BFCL-v4](#) introducing holistic agentic evaluation. Checkout [code](#) and [data](#).

Last Updated: 2025-12-16 [\[Change Log\]](#)

Search

Rank	Overall Acc	Model	Cost (\$)	Agentic		Multi Turn	Single Turn		Hallucination Measurement		Format Sensitivity		Latency (s)	Organization
				Web Search	Memory	Multi turn	Non-live (AST)	Live (AST)	Relevance	Irrelevance	Max Delta	SD		
				Overall Acc	Overall Acc	Overall Acc	Overall Acc	Overall Acc					Mean	
1	77.47	Claude-Opus-4-5-20251101 (FC)	86.55	84.5	73.76	68.38	88.58	79.79	62.5	84.72	N/A	N/A	4.38	Anthropic
2	73.24	Claude-Sonnet-4-5-20250929 (FC)	43.73	81	64.95	61.37	88.65	81.13	68.75	86.61	N/A	N/A	4.31	Anthropic
3	72.51	Gemini-3-Pro-Preview (Prompt)	298.47	80	61.72	60.75	90.65	83.12	68.75	85.59	8.5	1.7	12.08	Google
4	72.38	GLM-4.6 (FC thinking)	4.64	77.5	55.7	68	87.56	80.9	75	84.96	N/A	N/A	4.34	Zhipu AI
5	69.57	Grok-4-1-fast-reasoning (FC)	17.26	82.5	53.98	58.87	88.27	78.46	81.25	79.43	N/A	N/A	6.74	xAI
6	68.7	Claude-Haiku-4-5-20251001 (FC)	14.23	83.5	54.41	53.62	86.5	78.68	62.5	85.11	N/A	N/A	1.68	Anthropic

Stage 3: AI Technology



$$(80\%)^3 = 0.512$$

Exemplary multi-step agent

Stage 3: LLM Leaderboards

Arena Overview

Scroll to the right to see full stats of each model

First Place

Second Place

Third Place

Default

Compact View

Model	627 / 627	Overall	Expert	Hard Prompts	Coding	Math	Creative Writing	Instruction Following	Longer Query
claude-opus-4-6-thi_		1	2	1	1	2	1	1	1
claude-opus-4-6		2	1	2	2	3	5	2	2
gemini-3.1-pro-prev_		3	4	3	3	4	4	3	3
grok-4.20-beta1		4	22	5	6	17	2	10	13
gemini-3-pro		5	10	7	11	5	3	8	6
gpt-5.4-high		6	3	4	4	1	10	4	7
grok-4.20-beta-0309_		7	12	6	7	16	7	16	18
gpt-5.2-chat-latest_		8	16	8	8	12	15	11	15
gemini-3-flash		9	14	12	19	7	9	15	14
claude-opus-4-5-202_		10	7	9	5	8	6	5	4
grok-4.1-thinking		11	25	18	25	28	22	33	30
claude-opus-4-5-202_		12	8	11	12	13	8	7	8
gpt-5.4		13	6	14	20	9	31	9	10
qwen3.5-max-preview		14	15	17	16	10	11	17	21
gpt-5.3-chat-latest		15	34	13	13	25	35	22	17
gemini-3-flash (thi_		16	27	21	26	15	12	21	19
claude-sonnet-4-6		17	9	10	9	11	17	6	5
dola-seed-2.0-previ_		18	23	16	18	20	41	30	33
grok-4.1		19	39	26	30	45	23	31	29
glm-5		20	19	25	33	18	18	23	24
gpt-5.1-high		21	24	27	32	19	25	20	26
gpt-5.4-mini-high		22	13	22	24	-	48	19	22
kimi-k2.5-thinking		23	18	24	21	6	32	25	27
claude-sonnet-4-5-2_		24	11	15	10	21	20	12	11
claude-sonnet-4-5-2_		25	20	19	15	51	13	13	12

<https://huggingface.co/spaces/lmarena-ai/arena-leaderboard>

Stage 3: LLM Leaderboards

Independent analysis of AI

Understand the AI landscape to choose the best model and provider for your use case

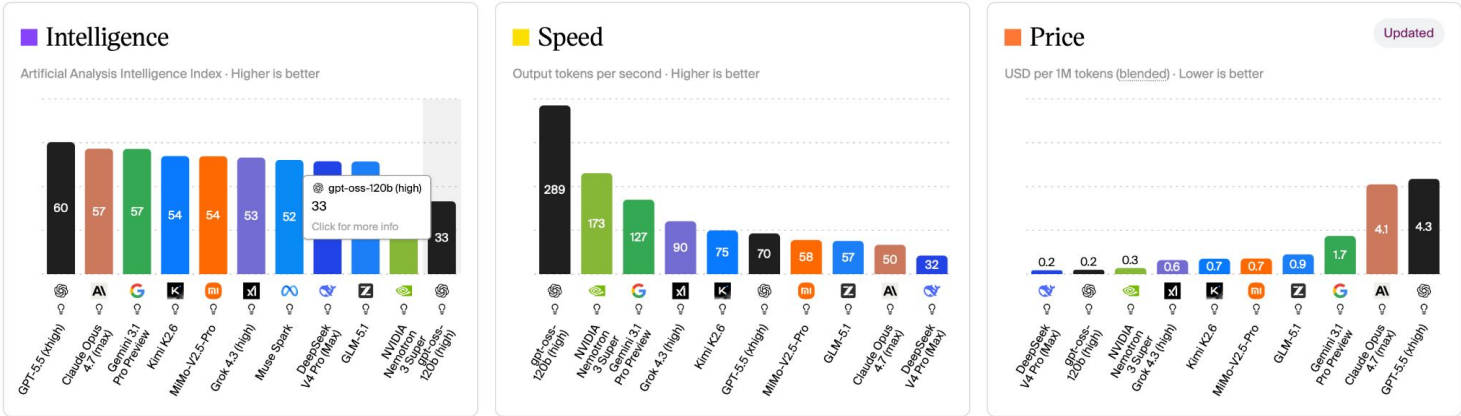
ANNOUNCEMENT

A new look for Artificial Analysis
Same mission, new look

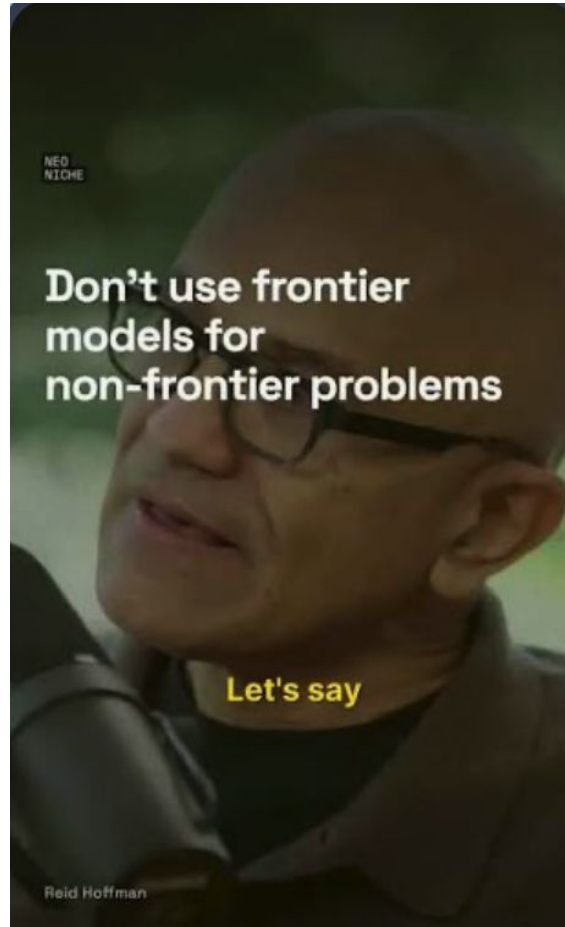
LAUNCH

Coding Agent Benchmarks
Introducing the Artificial Analysis Coding Agent Benchmarks

Highlights



Stage 3: Model is not the Moat



<https://www.instagram.com/reel/DaZYSFjkmnq/>

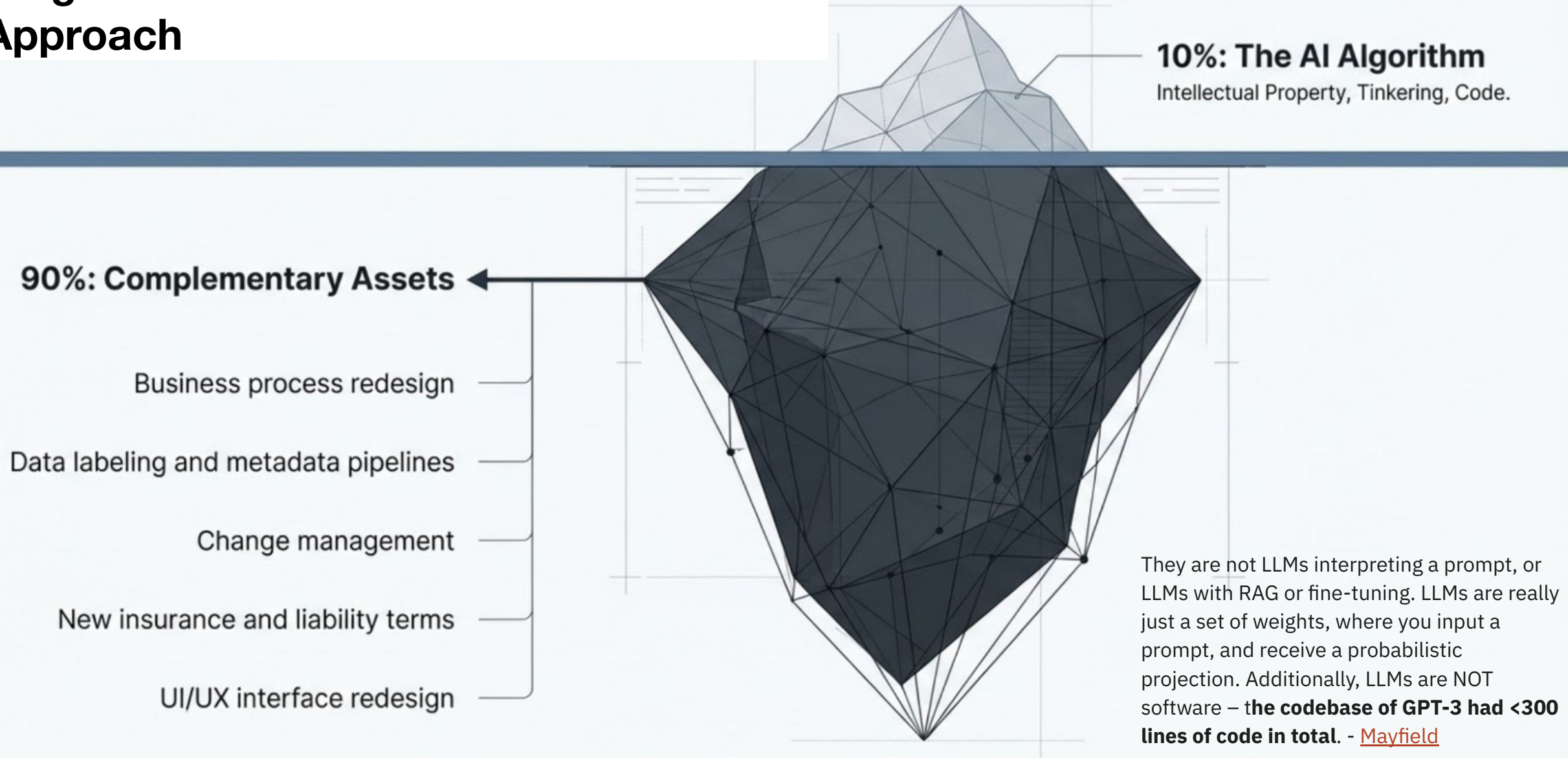
Stage 3: LLM intelligence levels

Agency Level	Description	Short name	Example Code
☆☆☆	LLM output has no impact on program flow	Simple processor	<code>process_llm_output(llm_response)</code>
★☆☆	LLM output controls an if/else switch	Router	<code>if llm_decision(): path_a() else: path_b()</code>
★★☆	LLM output controls function execution	Tool call	<code>run_function(llm_chosen_tool, llm_chosen_args)</code>
★★☆	LLM output controls iteration and program continuation	Multi-step Agent	<code>while llm_should_continue(): execute_next_step()</code>
★★★	One agentic workflow can start another agentic workflow	Multi-Agent	<code>if llm_trigger(): execute_agent()</code>
★★★	LLM acts in code, can define its own tools / start other agents	Code Agents	<code>def custom_tool(args): ...</code>

Source: https://huggingface.co/docs/smolagents/en/conceptual_guides/intro_agents

<https://medium.com/@dataenthusiast.io/ai-agents-when-your-problem-needs-one-and-when-it-doesnt-b0f200350df9>

Stage 4: Software & Business Process Approach



Software Engineering under the Iceberg of AI Products



Armand Ruiz · 2nd
VP of Product - AI Platform @IBM
1w · 🌐



Building AI agents is 5% AI and 100% software engineering.
You need:

1/ Smooth Integration: Seamlessly connect with existing systems.

2/ Coordinated Integration: Facilitate collaboration between agents and humans.

3/ Reliability: Deliver AI systems that perform consistently in production.

4/ High Scalability: Design for real-world loads.

5/ Constant Monitoring: Keep systems under continuous watch.

Your models & AI agents are only as good as the engineering that deploys them.



Sri Laasya Nutheti 🌐 @n_sri_laasya · Feb 14
Building AI agents is 5% AI and 100% software engineering



98

145

1.6K

133K



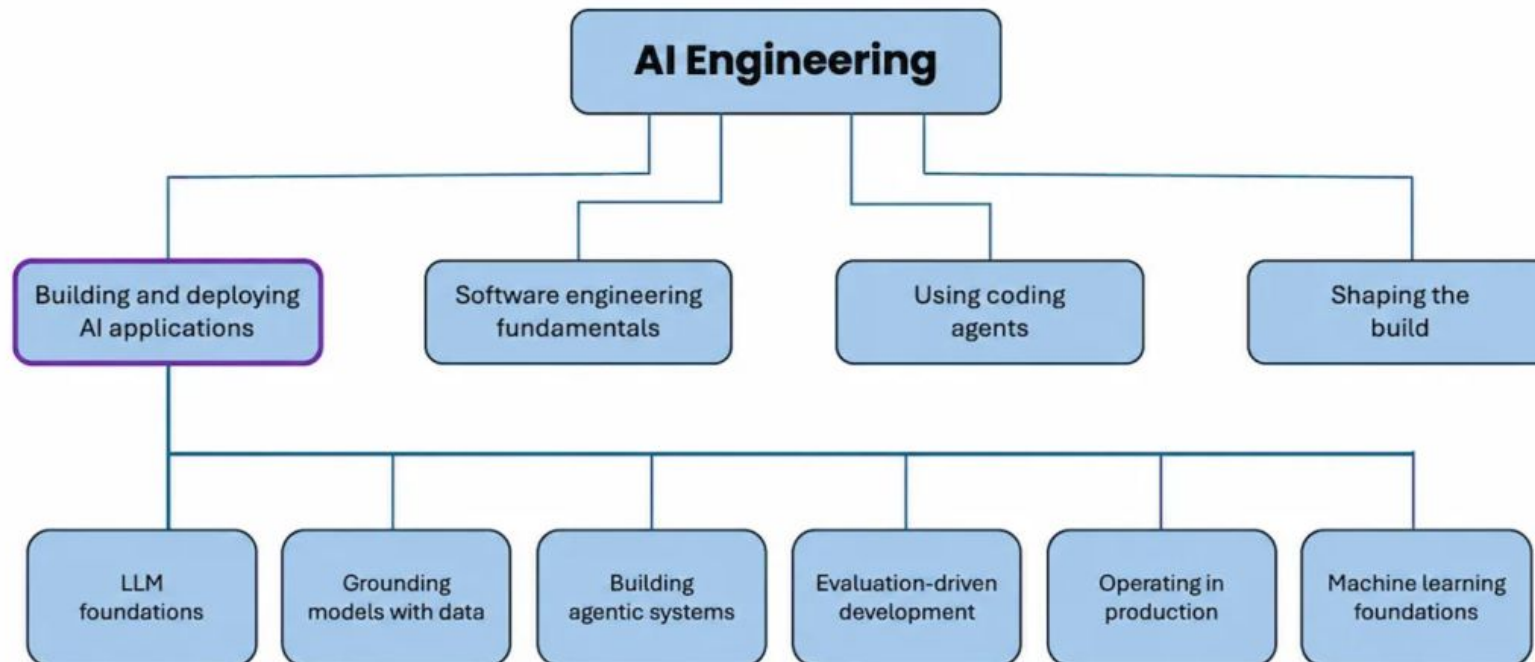
Software Engineering under the Iceberg of AI Products



Andrew Ng • Following
DeepLearning.AI, AI Fund and AI Aspire
1h •

The most important skills in Building and Deploying AI Applications.

AI Engineering Skills Map: Building and Deploying AI Applications



<https://www.deeplearning.ai/the-batch/the-ai-engineering-skills-map>

UX or AI revolution?

DATA-DRIVEN LEADERSHIP AND CAREERS

What's Different About Today's AI?

Hint: today's AI revolution isn't really about AI... it's about UX



Cassie Kozyrkov

Follow

4 min read · Apr 20, 2023



1.2K



21



Long before the generative AI art apps and chatbots splashed their way onto the tech scene, artificial intelligence was powering your online experiences and optimizing many of the businesses you enjoy offline. AI was everywhere, but it was hidden in plain sight as a subtle, unobtrusive component in software applications. If you didn't know how to look for the signs, it was easy to miss them. The subtlety was built-in — on purpose! — by user experience (UX) designers following a very intentional strategy.

AI was traditionally a subtle, unobtrusive component in software applications.

The Temptation Trap

One shot temptation:

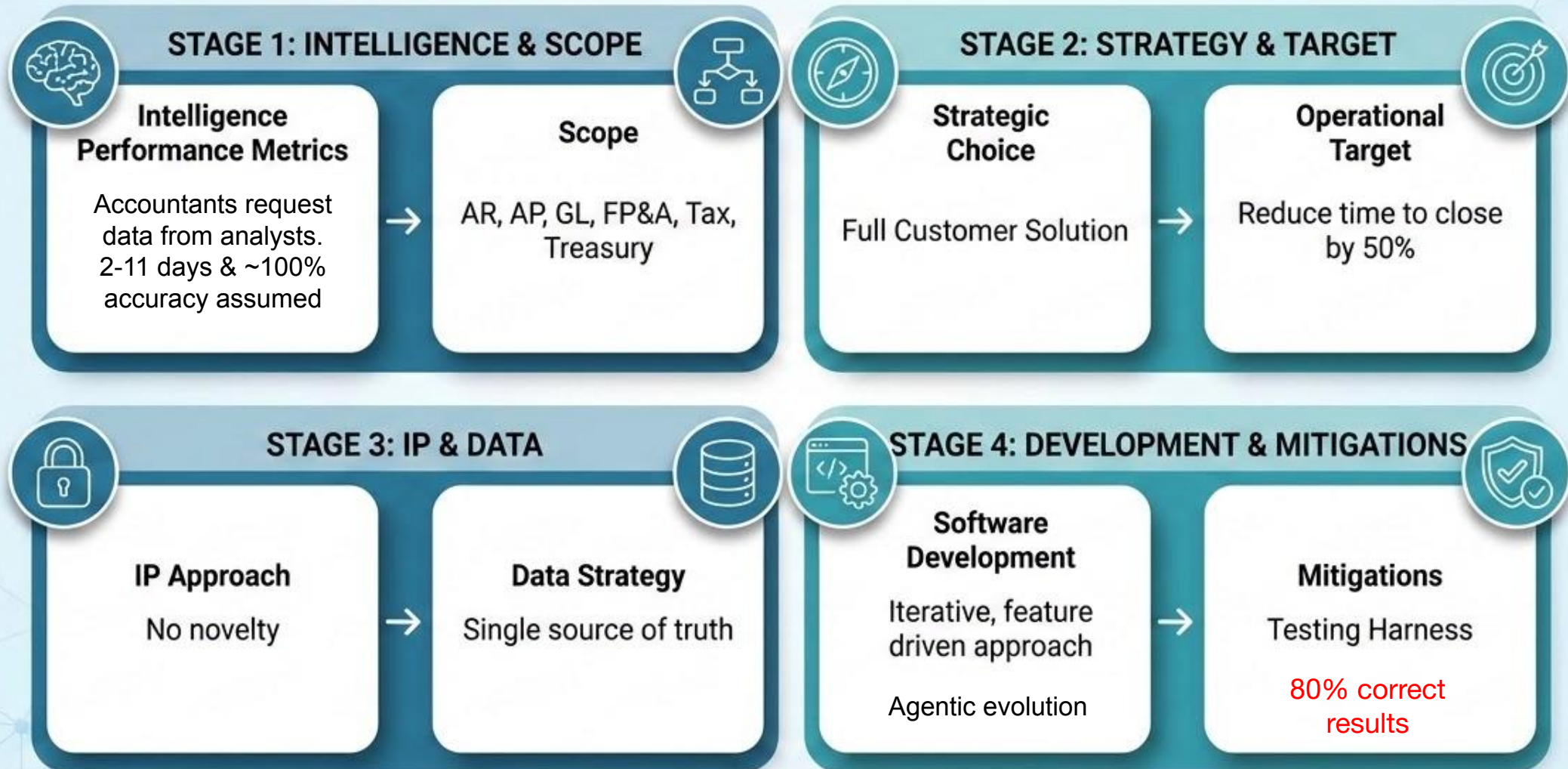
The false assumption that one LLM can simultaneously manage intent, data retrieval, and complex logic.

- **Context Decay:** Overloading the prompt window causes the model to "lose the thread" of the original user objective.
- **Forced Hallucination:** Pushing a single model to act as both researcher and executor triggers confident, fabricated outputs.
- **Architectural Unreliability:** Without modularity, the system inevitably collapses under "Context Rot" and logic breakdowns.

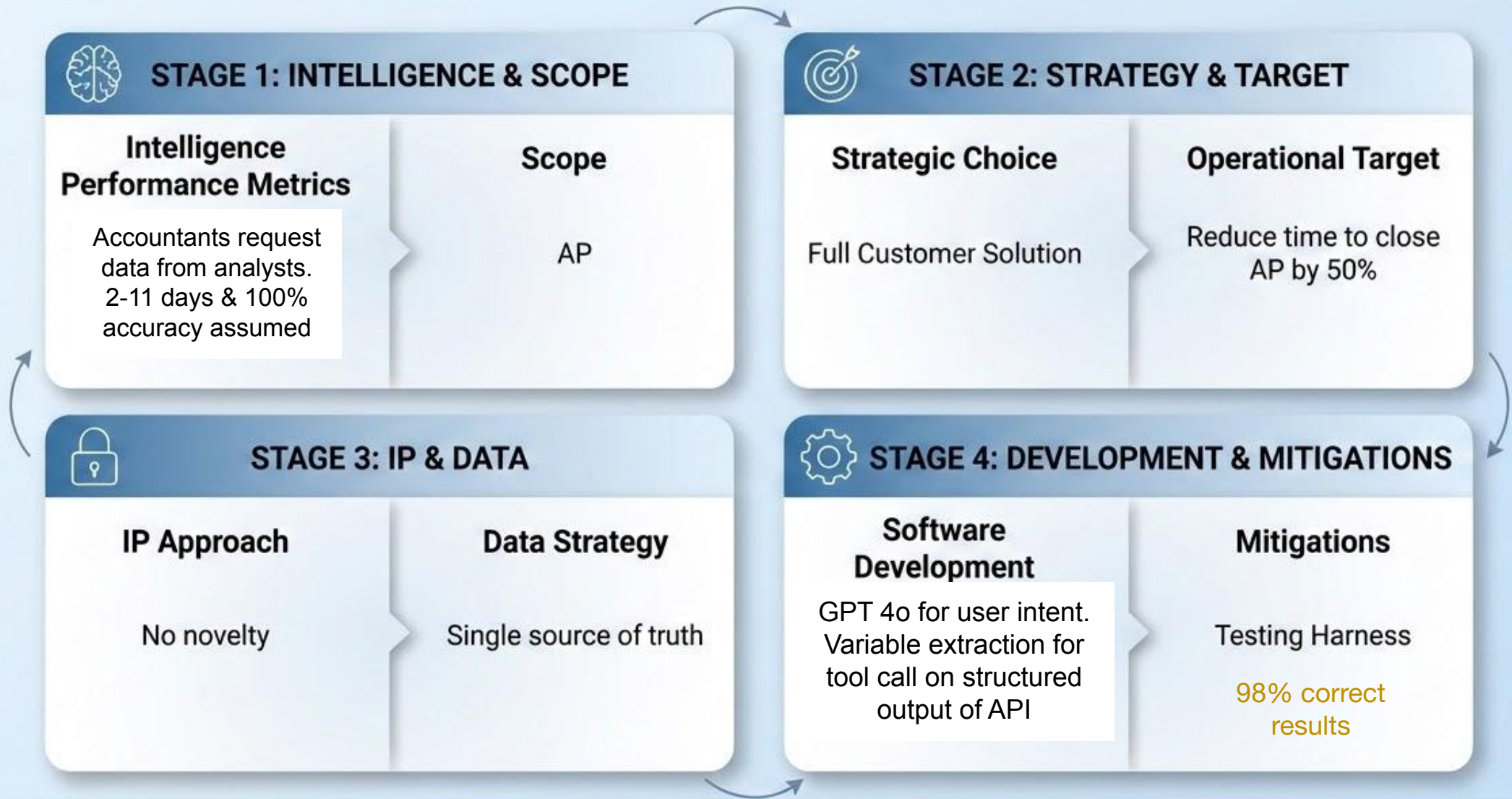


Case Study: How might we reduce time & effort to close books and deliver financial reports using AI?

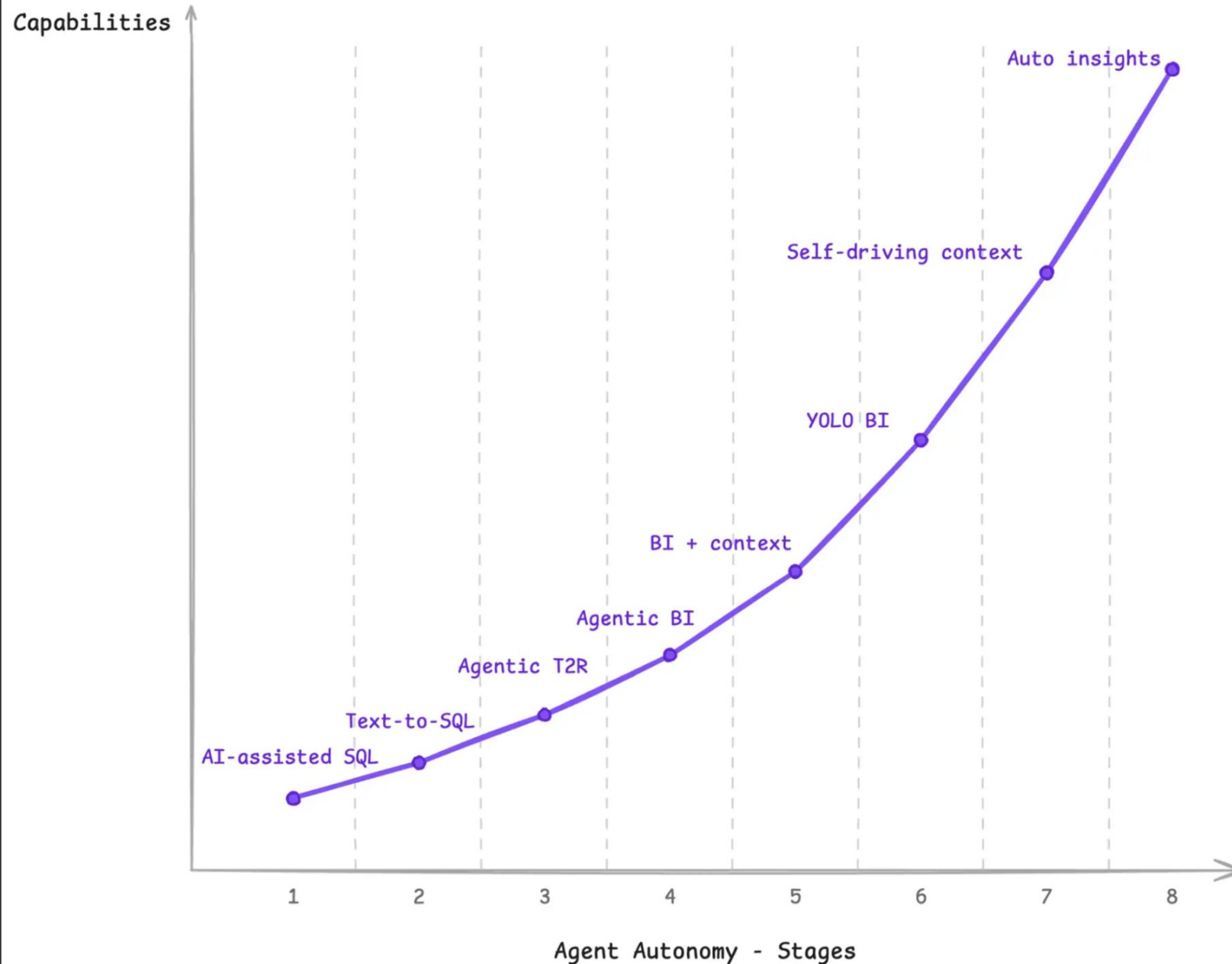
Case Study: Building AI agent for Finance & Accounting. Iteration 1



Case Study: Building AI agent for Finance & Accounting. Iteration 2



Agentic AI stages for analysts



https://motherduck.com/blog/water-town-agent-swarm-data-stack/?utm_campaign=Blog%20Posts&utm_medium=email&hsenc=p2ANqtz-9zX8DG6FplQ9F5Ulf6G804bS86qkCDXy0divBMrHRXcTMOpoXpnH1MuknNQOXbp5_FThyPIU2NfBpDxZNRs9d9QzS5Qg&hsmi=413953912&utm_content=413953912&utm_source=hs_email

Interpretability and observability are key!

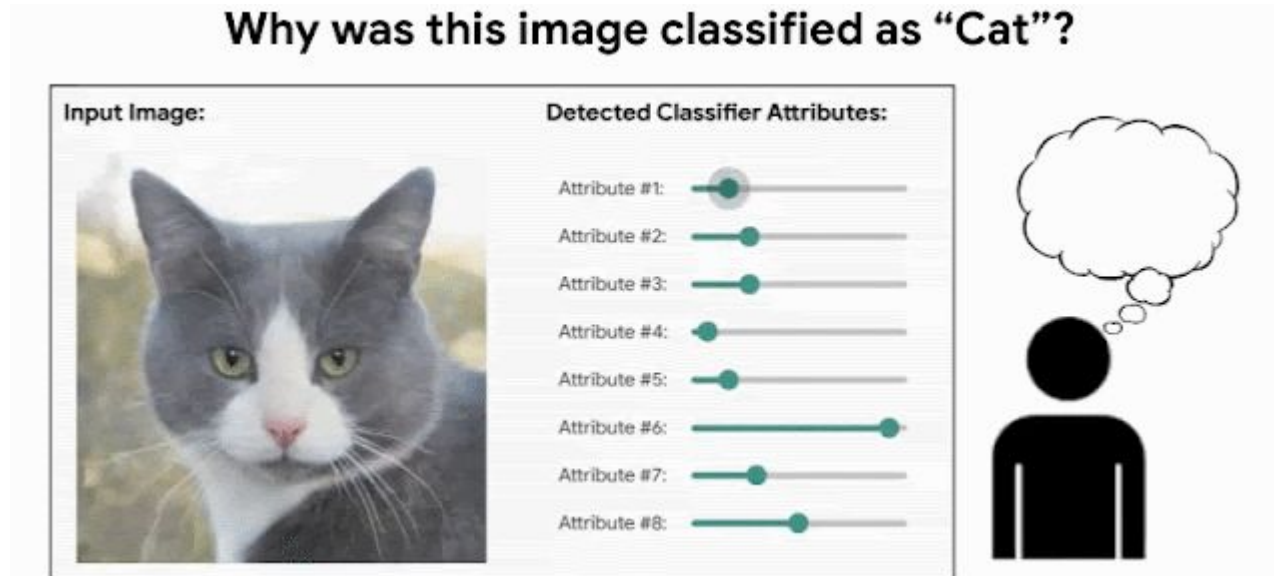
Classifying a Dog vs Wolf?



Classifying between a Husky and a Wolf is not a straight forward task, yet algorithms manage to obtain 90% accuracy. Or do they?

<https://becominghuman.ai/its-magic-i-owe-you-no-explanation-explainableai-43e798273a08>

Classifying a Dog vs Wolf?



Explaining a Cat vs. Dog Classifier: StyleX provides the top-K discovered disentangled attributes which explain the classification. Moving each knob manipulates only the corresponding attribute in the image, keeping other attributes of the subject fixed.

<https://ai.googleblog.com/2022/01/introducing-stylex-new-approach-for.html>

<https://cloud.google.com/vertex-ai/docs/explainable-ai/overview>

Agent Trace

Arize AX

inventions

Home

Observe

Projects

Monitors

Dashboards

Develop

Datasets & Experiments

Prompt Playground

Prompt Hub

Evaluate

Evals & Tasks

Annotation Configs

Labeling Queues

Settings

Invite User

APIs

Help

Sanjana Yeddula

travel-agent-demo

ID

+ New Dashboard

+ New Monitor

+ New Eval Task

...

Traces

Spans

Sessions

Agent Graph

Agent Path

Oct 1 - Oct 13

Live

Filter by: "span property" = 'dimension'

demo-save

sanjana2

tools

error

essential info tool

budget tool span

local flavor

+

Total Traces 51

Total Spans 465

Total Tokens 123.1k

Total Cost \$0.386092

Latency P50 1.35s

Latency P99 38.28s

51 traces

Download

Refresh

Reset

Columns

	Status	Kind	Name	Input	Output	Start Time	Latency	Total Tokens	Evaluations
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:47 ...	23.02s	2415	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:46 ...	22.56s	2180	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:43 ...	18.44s	2372	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:42 ...	14.81s	2237	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:41 PM	14.97s	2305	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:40 ...	17.78s	2283	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:37 ...	17.83s	2247	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:36 ...	27.39s	2445	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:34 ...	23.14s	2293	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:31 PM	21.49s	2301	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:28 ...	22.14s	2318	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:27 ...	20.58s	2220	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:25 ...	20.67s	2271	
☐	✓	AGENT	TripPl...	{"run_response":...		10/13/2025, 02:24 ...	14.43s	2220	

Ghosts in the machines

Diagnosing and mitigating AI's inherent limitations.

Risk & Mitigation

Limitation (Cancer)	Description	Actionable Mitigation
1. Adversarial Attacks	Small, strategic modifications to input data that fool algorithms.	Robust adversarial training. ✓
2. Lack of Generalization	Inability of models to perform outside narrow training parameters.	Diverse dataset curation. ✓
3. Bias	Reflection of historical prejudices present in training data.	Algorithmic auditing. ✓
4. Explainability	Black box deep learning decisions lacking transparency.	Integration of Chain-of-Thought prompting. ✓
5. Unintended Behavior	Unexpected outputs undermining the system's intended goal.	Human-in-the-loop oversight. ✓

EMCODE - Module 2 primer. 10 Notebooks total for the course

Designing and Building AI Products an... > Assignment 2.1-Training and Testing



File View Nbgrader Settings Help

Files Running

Open Download Rename Duplicate Move to Trash

Filter New

/

Name		Modified
☐	data	9 days ago
☒	Assignment 2.1-Training and Testing.ipynb	8 minutes ago

https://classroom.emeritus.org/courses/13615/assignments/318860?module_item_id=2450501



Q&A

Stage 1: Intelligence Performance Metrics

Feature	Raw API Inference (The "Engine")	Agentic Workflow (The "Employee")
How it Works	Single-pass prediction (Prompt → Output).	Multi-step loop (Reason → Act → Reflect).
Speed	Milliseconds (Instantaneous).	Seconds to Minutes (Slower execution).
Cost Profile	Extremely Low (1 Input + 1 Output).	High (Requires 5-10+ hidden API loops).
Reliability	Highly Predictable. Engineered to output strict data formats (e.g., Structured JSON).	Variable. Agents can occasionally get stuck in loops.
Ideal Target	High-volume, structured, repetitive scale.	Low-volume, ambiguous, high-value problem solving.
Enterprise Examples	• Extracting data from 10,000 PDFs. • Categorizing daily IT support tickets. • Real-time chat moderation.	• Researching & drafting custom client briefs. • Autonomously debugging server outages. • Resolving complex customer complaints.

Stage 1: Performance Metrics - AI > LLM



<https://www.youtube.com/shorts/Lt1A60zHv6A>

Case Study: Building AI agent for Finance & Accounting. Iteration 1

Stage 1: Intelligence Performance Metrics

Accountants request
data from analysts. 2-11
days & 100% accuracy
assumed

Stage 1: Scope

AP, AR, GL, FP&A, Tax,
Treasury

Stage 2: Strategic Choice

Full Customer Solution

Stage 2: Operational Target

Reduce time to close by
50%

Stage 3: IP Approach

No novelty, PII
protection

Stage 3: Data Strategy

Single source of truth

Stage 4: Software Development

Iterative, feature driven
approach

Stage 4: Mitigations

Testing Harness